

wrote the best-selling book, *Beat the Dealer*.⁴ It is, even today, regarded as a classic work, and blackjack players the world over rely on Basic Strategy to optimize their card play.

In the 1960s, casinos offered single-deck blackjack and dealt the entire deck. Thorp calculated that players who counted cards and scaled their bets based on the residual cards left in the deck had an edge over the casinos. He used the Kelly Formula to figure out how much of your bankroll you ought to bet each time based on how favorable the odds were. For example, if the deck had an overrepresentation of tens and aces, that was good for the player. If the odds were 52:48 in the favor of the player, the Kelly Formula suggested that the player bet 4 percent of his bankroll. That's what Thorp would endeavor to do with every hand.

For Thorp, this wasn't an academic exercise. He started frequenting the Nevada casinos and cleaned up. The casinos didn't understand why he was consistently winning, but, with the mob running the casinos, they didn't wait to understand. They simply showed him the door and made it very clear that if he ever returned, the reception wouldn't be so civil.

When Thorp published *Beat the Dealer*,⁵ players the world over started cleaning up. Casino owners also read Thorp's book and began to make changes to the game. Over the past four decades, the game has gone through numerous changes. Each time the casinos made a change, some smart gambler would figure out a way to beat the system. Then the casinos would figure it out and make another change. Today, most casinos deal from a shoe of six to eight decks. They don't play the last couple of decks and pit bosses watch the action like hawks. In some casinos, auto shufflers recycle the used cards back in real time—ensuring that the card pool never has an over- or underrepresentation of any specific cards.

Thorp reflected on this changing reality (along with the onerous threats) and decided that he'd be far better off if he applied his talents to a casino where:

There were no table limits.

The offered odds were vastly better.